

AMFIINTEL TERMINAL - Project Documentation

Enterprise Mutual Fund Intelligence & Portfolio Analytics Platform

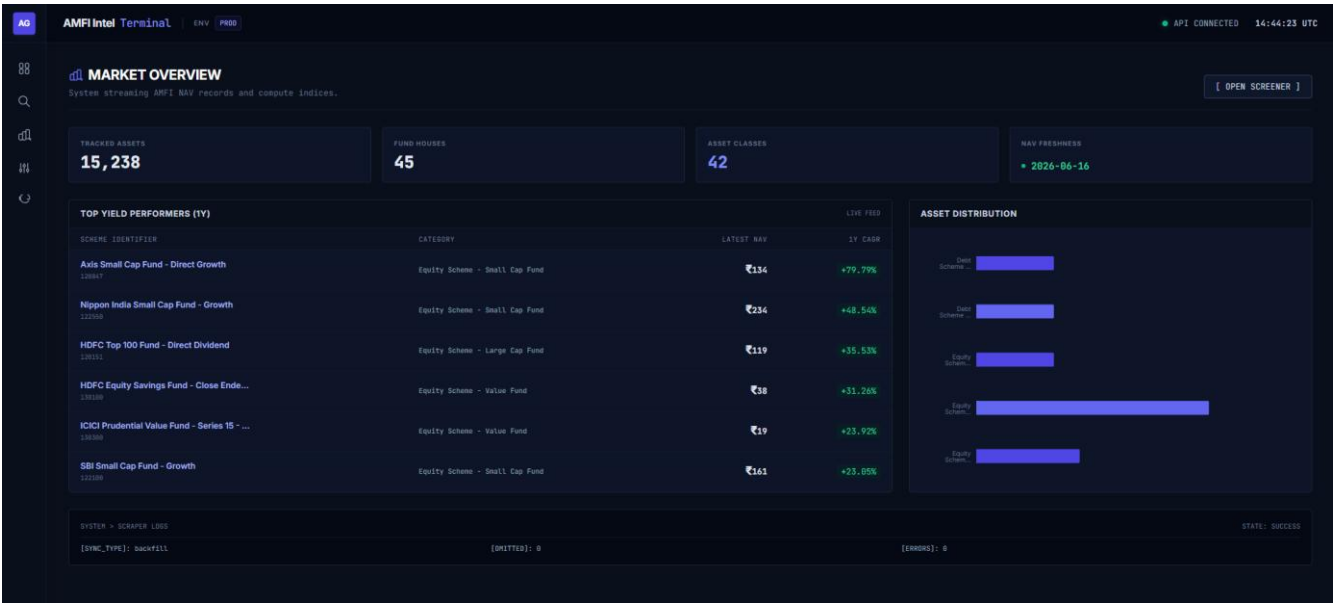
Abstract

AMFIIntel Terminal is a production-grade Mutual Fund Intelligence and Portfolio Analytics Platform that transforms publicly available AMFI data into actionable investment insights. The platform automates data collection, historical NAV tracking, analytics generation, risk assessment, and portfolio analysis through a centralized fintech dashboard, enabling users to conduct comprehensive mutual fund research from a single interface.

Built using Next.js, FastAPI, PostgreSQL, and automated data engineering pipelines, the platform provides advanced capabilities such as fund comparison, portfolio overlap detection, diversification analysis, historical trend visualization, and professional report generation. By combining financial analytics with modern visualization and scalable backend architecture, AMFIIntel delivers a complete investment research ecosystem for investors, analysts, and financial professionals.

Introduction

AMFIIntel Terminal is a production-grade Mutual Fund Intelligence and Portfolio Analytics Platform designed to transform publicly available AMFI data into actionable investment insights. The platform automates data collection, historical NAV tracking, financial analytics, portfolio overlap detection, risk assessment, and report generation through a unified fintech dashboard. By combining data engineering, backend services, analytical processing, and modern visualization techniques, AMFIIntel enables investors and analysts to conduct comprehensive mutual fund research, performance evaluation, and portfolio analysis from a single platform.



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Mutual Fund Asset Explorer

Perform queries across historical Indian mutual funds, filter by category class, and analyze performance CAGR records.

LATEST NAV EXPLORER

NAV DOWNLOAD & HISTORY

AUM DISCLOSURES

Download Daily NAV Reports

Download plain-text reports directly from AMFI India for all classifications.

TXT

Complete NAV Report

Includes all active schemes (NAVAll.txt)

Download Report

TXT

Open Ended NAV Report

Equity, hybrid & debt schemes (NAVOpen.txt)

Download Report

TXT

Close Ended NAV Report

Closed schemes and ETFs (NAVClose.txt)

Download Report

TXT

Interval Fund NAV Report

Special interval term schemes (NAVInterval.txt)

Download Report

Historical NAV Query Engine

Specify a target Asset Management Company, scheme class, and dates. Queries the portal dynamically.

MUTUAL FUND HOUSE

SCHEME CLASS TYPE

FROM DATE

TO DATE

Select Mutual Fund (All)

Select Type (All)

dd-mm-yyyy

dd-mm-yyyy

Maximum query range is restricted to 90 days.

Get Report

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☒ SBI Bluechip Fund - Direct Dividend

Equity Scheme - Large Cap Fund

☐ SBI Liquid Fund - Growth

Debt Scheme - Liquid Fund

☐ SBI Nifty 50 Index Fund - Growth

Other Schemes - Index Funds

☐ SBI Tax Advantage Fund - Series III - Growth

Equity Scheme - Value Fund

☐ SBI Bluechip Fund - Direct Growth

Equity Scheme - Large Cap Fund

☒ SBI Magnum Midcap Fund - Growth

Equity Scheme - Mid Cap Fund

☐ SBI Nifty Index Fund - Growth

Equity Scheme - Large Cap Fund

☐ SBI Healthcare Opportunities Fund - Growth

Equity Scheme - Sectoral Thematic

☐ SBI Monthly Interval Fund - Plan A - Growth

Debt Scheme - Interval Fund

☒ SBI Small Cap Fund - Growth

Equity Scheme - Small Cap Fund

₹10,000 COMPOUND INVESTMENT GROWTH SERIES

SBI Magnum Midcap Fund - Growth

Axis Small Cap Fund - Direct Dividend

SBI Small Cap Fund - Growth

SBI Bluechip Fund - Direct Dividend

₹18000

₹16000

₹14000

₹12000

₹10000

₹8000

₹6000

₹4000

₹2000

₹0

Jun 25

Jul 25

Jul 25

Aug 25

Aug 25

Sep 25

Sep 25

Oct 25

Oct 25

Nov 25

Nov 25

Dec 25

Dec 25

Jan 26

Jan 26

Feb 26

Feb 26

Mar 26

Mar 26

Apr 26

Apr 26

May 26

May 26

RISK & RETURN COMPARISONS

Axis Small Cap Fund - Direct Dividend

1Y Returns: +26.54%

Volatility: +28.76%

Sharpe Ratio: 8.23

Max DD: -48.51%

SBI Bluechip Fund - Direct Dividend

1Y Returns: -8.14%

Volatility: +13.51%

Sharpe Ratio: 8.97

Max DD: -18.84%

SBI Magnum Midcap Fund - Growth

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Portfolio Lab

Simulate combinations of mutual funds to detect holdings overlap, sector concentration, and overall risk diversification before investing.

Simulate Holdings

Selected Funds

Search and add fund...

SBI Bluechip Fund - Direct Dividend

Allocation: 25%

HDFC Flexi Cap Fund - Growth

Allocation: 30%

Axis Small Cap Fund - Direct Growth

Allocation: 45%

Portfolio Overlap Intelligence

OVERLAP RISK

30.1%

DIVERSIFICATION SCORE

69.9/100

COMMON HOLDINGS

TCS

Infosys

Hindustan Unilever

L&T

SECTOR EXPOSURE

Construction

FMCG

Financial Services

IT

Telecommunication

Problem Statement

The Indian mutual fund ecosystem comprises thousands of schemes across multiple Asset Management Companies (AMCs), categories, and investment strategies. Although data is publicly available through AMFI, AMC websites, and financial research portals, it remains fragmented and difficult to analyze efficiently. Investors often spend significant time gathering information from multiple sources, resulting in inconsistent research processes and limited visibility into fund performance, risk, and market trends.

Additionally, most platforms focus on displaying current fund data while offering limited historical analysis, portfolio intelligence, and reporting capabilities. Investors struggle to evaluate long-term performance, compare funds across key metrics, identify portfolio overlaps, and assess diversification risks. The absence of integrated analytics and research tools makes informed investment decision-making both complex and time-consuming.

Project Vision

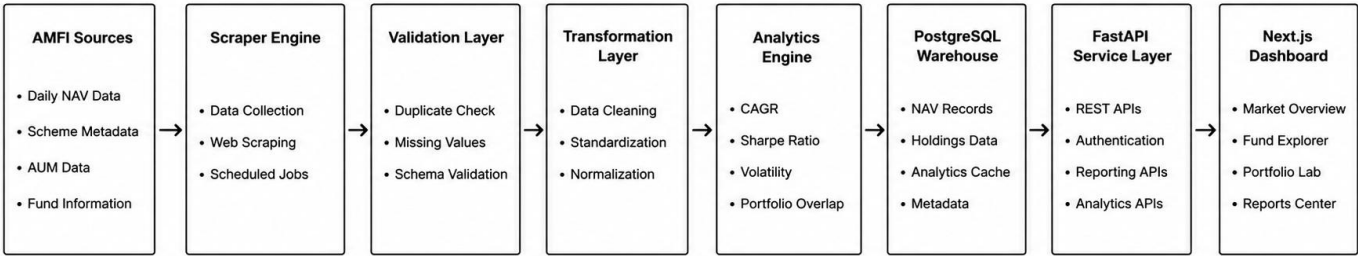
The vision of AMFIIntel Terminal is to create a comprehensive Mutual Fund Intelligence Platform that transforms fragmented financial data into meaningful, actionable investment insights. The platform aims to bridge the gap between raw mutual fund information and informed decision-making by providing investors, analysts, and researchers with a centralized ecosystem for fund discovery, performance evaluation, risk assessment, and portfolio analysis.

To achieve this vision, the platform focuses on:

- **Automated Data Intelligence through continuous synchronization and processing of AMFI mutual fund datasets.**
- **Historical Data Warehousing for maintaining and analyzing long-term NAV records, fund metadata, and market trends.**
- **Advanced Financial Analytics including CAGR, rolling returns, volatility metrics, risk scores, Sharpe ratios, and performance benchmarking.**
- **Portfolio Intelligence & Risk Assessment through holdings overlap detection, diversification analysis, concentration risk evaluation, and sector exposure insights.**
- **Professional Reporting & Data Accessibility through automated generation of CSV, Excel, and PDF reports for research, analysis, and decision support.**

Ultimately, AMFIIntel Terminal aspires to evolve beyond a traditional mutual fund dashboard into a scalable fintech intelligence platform that empowers users to conduct comprehensive investment research, uncover portfolio risks, and make informed financial decisions with confidence.

System Architecture



Technology Stack

Frontend

Next.js

Used for:

- Server-side rendering
- Routing

TypeScript

Provides:

- Type safety

Tailwind CSS

Used for:

- Consistent styling
- Rapid UI development

ShadCN UI

Provides:

- Accessible interfaces
- Enterprise design system

Recharts

Used for:

- NAV visualizations
- Performance analytics

Backend

FastAPI

Acts as the central API gateway.

Responsible for:

- Authentication
 - Fund APIs
 - Analytics APIs
 - Reporting APIs
 - Portfolio APIs
-

SQLAlchemy

Used as ORM layer.

Provides:

- Database abstraction
 - Query optimization
 - Relationship management
-

APScheduler

Handles:

- Daily synchronization jobs
 - Historical backfills
 - Automated refreshes
-

Database Layer

PostgreSQL

Used as the primary data warehouse.

Stores:

Fund Metadata

- AMC Information

- Scheme Information
- Categories
- Benchmarks

Historical NAV Records

Daily NAV history for all schemes.

Asset Allocation

Fund allocation across:

- Equity
- Debt
- Cash
- Gold

Holdings Repository

Stores:

- Company Holdings
- Sector Data
- Allocation Percentages

Analytics Cache

Stores:

- Rankings
- CAGR Metrics
- Risk Scores

AMFI Intel Terminal

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Scrapper & Data Pipeline Auditing

Monitor daily ingestion jobs, audit data parsing consistency, and view scheduler job status logs.

LATEST EXECUTION AUDIT TRAIL

EXECUTION ID	EXECUTION CLASS
Run #1	Backfill Ingestion
START BOUNDARY	COMPLETED BOUNDARY
22 May 2026	22 May 2026
RECORDS UPBETED	OMITTED ROWS
+29792 POWS	0 POWS
OMITTED ERRORS	EXECUTION STATUS
0 POWS	Success

SCHEDULER CLI BACKFILLS

Use the built-in click CLI hooks inside the virtual environment terminal to manually backfill historical date ranges.

RUN DAILY SYNC

python -s scheduler.cli run-now

BULK HISTORICAL BACKFILL

python -s scheduler.cli backfill --from 2023-01-01 --to 2023-12-31

Data Engineering Pipeline

Step 1 — Data Extraction

The scraper collects:

- Scheme metadata
 - Daily NAV records
 - Historical NAV datasets
 - AUM disclosures
-

Step 2 — Validation

The system validates:

- Missing values
- Invalid records
- Duplicate entries

before processing.

Step 3 — Transformation

Raw AMFI data is normalized into a structured format.

This includes:

- Date standardization
 - Category mapping
 - Numeric formatting
-

Step 4 — Analytics Generation

The analytics engine computes:

CAGR

Measures long-term annualized growth.

Rolling Returns

Evaluates consistency over time.

Volatility

Measures return fluctuations.

Sharpe Ratio

Evaluates risk-adjusted performance.

Drawdown

Measures downside exposure.

Step 5 — Database Synchronization

Validated records are inserted into PostgreSQL using optimized bulk operations.

Dashboard Modules

Module 1 — Market Overview

Provides a real-time snapshot of the mutual fund ecosystem.

Displays:

- Tracked Assets
- Fund Houses
- Asset Classes
- NAV Freshness
- Top Yield Performers

Purpose:

Enable users to quickly understand current market conditions.

Module 2 — Mutual Fund Asset Explorer

Core research workspace.

Supports:

- Fund Discovery
- Search
- Filtering
- Category Analysis

- Risk Evaluation

Users can explore thousands of mutual funds through a structured interface.

Module 3 — NAV Query Engine

Provides direct access to historical NAV records.

Supports:

- Date Range Queries
- AMC Filters
- Category Filters

Enables detailed historical analysis.

Module 4 — AUM Intelligence Dashboard

Analyzes Assets Under Management across:

- Fund Categories
- Fund Houses
- Direct vs Regular Plans

Provides market-wide capital allocation insights.

Module 5 — Fund Comparison Engine

Allows users to compare multiple funds simultaneously.

Comparison Metrics:

- CAGR
- NAV Growth
- Volatility
- Sharpe Ratio
- Drawdown

Provides side-by-side investment evaluation.

Module 6 — Portfolio Lab

One of the flagship features.

Users can simulate mutual fund combinations.

The system computes:

Overlap Risk

Detects common holdings.

Diversification Score

Measures portfolio diversity.

Sector Exposure

Analyzes allocation concentration.

Concentration Risk

Identifies hidden portfolio vulnerabilities.

Module 7 — Pipeline Auditing Console

Provides transparency into data ingestion operations.

Tracks:

- Scheduler Jobs
- Backfills
- Records Processed
- Errors
- Execution Logs

This module improves operational reliability and monitoring.

Reporting Engine

The reporting system generates:

CSV Exports

Raw analytical datasets.

Excel Reports

Structured spreadsheet reports.

PDF Reports

Investor-grade research documents.

Supported Reports:

- Fund Reports
 - Portfolio Reports
 - Comparison Reports
 - Historical Trend Reports
-

Security & Authentication

Implemented using:

JWT Authentication

Provides secure API access.

Password Hashing

Protects user credentials.

Role-Based Access

Supports:

- Admin
- Analyst
- Investor

roles.

Key Achievements

- Automated collection of AMFI mutual fund data.
- Built a complete financial data engineering pipeline.
- Designed a scalable PostgreSQL analytical warehouse.
- Developed a FastAPI-based service architecture.
- Implemented portfolio overlap intelligence.
- Created a Bloomberg-inspired fintech dashboard.
- Developed professional reporting capabilities.
- Enabled historical research and comparative analytics.

Conclusion

AMFIIntel Terminal transforms publicly available mutual fund information into a comprehensive investment intelligence ecosystem. By integrating automated data engineering, financial analytics, portfolio intelligence, historical research capabilities, and modern visualization technologies, the platform provides investors and analysts with a powerful environment for conducting data-driven mutual fund research and portfolio evaluation.

The project demonstrates expertise in full-stack development, data engineering, backend architecture, financial analytics, database design, system automation, and modern dashboard engineering, making it a production-grade fintech application rather than a conventional academic project.